

Debadrita Baksi

Kolkata, West Bengal

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Professional Summary

B.Tech CSE student, 3x hackathon dominator, and UI/UX frontend architect specializing in high-performance digital ecosystems. Experienced in engineering enterprise-grade AI forensic dashboards, decentralized IoT navigation interfaces, and multi-agent intelligence platforms. Proficient in translating complex system logic into intuitive user experiences, bridging interactive React and WebGL frontends with robust Python and C-based backend infrastructure to solve real-world engineering challenges.

Skills & Achievements

- **Core Languages:** HTML5, CSS3, Java, Python, C
- **Familiarity & Tools:** JavaScript, React.js, Three.js, Git, GitHub, MySQL
- **Core Concepts:** UI/UX Design, Web Page Structuring, Object-Oriented Programming (OOP)
- **Achievements:** 1st place - App Development Competition [App-E-Teaser] (JISTech 2K26),
2nd Runner-Up - Theme Based Project [AI Enabled System] (JISTech 2K26),
3rd Runner-Up - Crazy Build 2K26,
Best Beginners Track Winner - HexaFalls 2 Hackathon (HexaFalls 2 Techfest 2026)

Top Projects

Maargdarshan | AI-Enabled Smart Navigation Ecosystem

*React.js, JavaScript, Python,
HTML5, CSS3, Three.js, FastAPI*

(Team Project - Frontend Web Developer, UI/UX Designer & Backend Co-developer)

Live Prototypes: Website | App (*Prototypes in active Beta testing)

- **UI/UX & Interactive Styling:** Designed responsive UI components and thematic CSS styling, integrating fluid physics (satin/water effects) and custom interactive motifs (e.g., dynamic peacock feather cursor) for an engaging user experience.
- **Frontend Layout & Design:** Wireframed and co-developed the frontend visual architecture for the real-time caretaker web dashboard of a decentralized IoT navigation ecosystem.
- **Backend & Cloud Integration:** Engineered the Python-based server architecture (FastAPI) and cloud APIs, handling real-time software integration, secure data routing, and telemetry synchronization between the ESP32 hardware and the web dashboard.
- **Digital Twin Integration:** Assisted in integrating the WebGL 3D hardware mirroring interface into the dashboard to help visualize the physical orientation of the remote hardware.

Chanakya-AI | Real-Time Competitive Intelligence Engine

Next.js, React, Tailwind CSS

[Team Project - Lead Frontend Engineer (Dashboard) & UI/UX Developer]

- Engineered the complex frontend dashboard and Hacker Theatre UI for a multi-agent AI platform, translating overarching design blueprints into functional software.
- Implemented advanced glassmorphic state components and staggered CSS animation loops to visualize heavy server processing without blocking the client's main thread.
- Developed the interactive Self-Strategy data matrices and the real-time Global Context Chat interface, ensuring seamless state management and responsive rendering.

Patronus | Multimodal AI Forensics Ecosystem

React, Tailwind CSS, WebSockets, Zustand

[Team Project - Frontend Co-Developer & UI/UX Architect]

- **Enterprise UI/UX Architecture:** Architected the comprehensive visual infrastructure across all 5 forensic modules (Video, Audio, Image, Text, and Document), engineering a strict, authoritative "cyber-forensics" dark-mode interface utilizing Tailwind CSS to meet enterprise standards.
- **Real-Time DOM Rendering:** Engineered a highly resilient React frontend capable of intercepting and parsing high-frequency backend WebSocket payloads, ensuring the flawless 60FPS rendering of live Base64 Laplacian edge-maps, Grad-CAM heatmaps, and streaming AI verdicts.
- **Global State & Memory Management:** Implemented a robust, anti-fragile global state architecture using Zustand to eliminate prop-drilling and safely synchronize complex, asynchronous AI payloads, preventing UI freezing and memory leaks during heavy data ingestion.

Education

B.Tech in Computer Science and Engineering (2025-2029)

JIS College of Engineering, Kalyani

Pursuing

Interests

- Interactive 3D Web Rendering & Digital Asset Creation
- Backend Logic Design & Full-Stack Architecture
- Competitive Hackathon Sprints & Tech Community Building